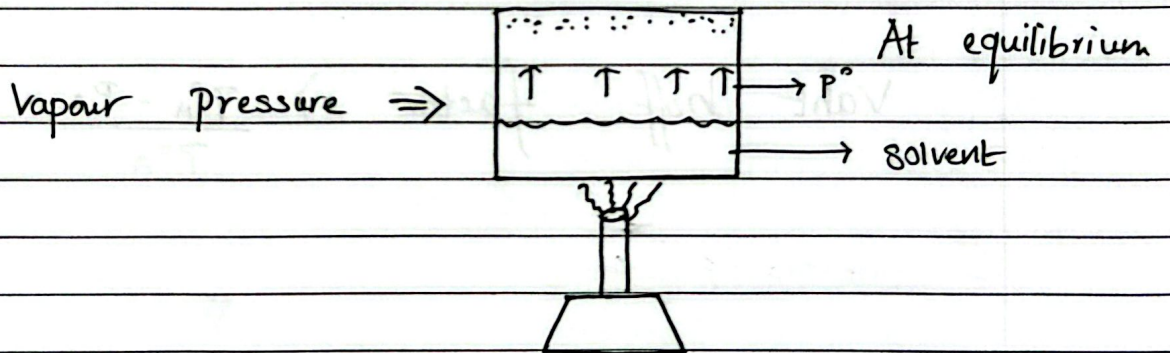


Solutions

— Colligative Properties depends on number of moles of solute

1. Relative lowering of Vapour Pressure.
2. Elevation in Boiling point.
3. Depression in Freezing Point.
4. Osmotic Pressure.



=> whenever a non-volatile solute is added to solvent, its vapour pressure will decrease

=> Raoult's Law

For a solution of volatile liquids, the partial vapour pressure is

$$\text{Total pressure} = P_A^0 x_A + P_B^0 x_B$$

$$P_i \propto x_i$$

$$\frac{P^{\circ}_A - P_A}{P^{\circ}_A} = x_B.$$

or

$$\frac{P^{\circ}_A - P_A}{P^{\circ}_A} = \frac{n_B}{n_A + n_B}.$$

or

$$\frac{P^{\circ}_A - P_A}{P^{\circ}_A} = \frac{W_2 M_1}{W_1 M_2}$$

P°_A = Pure non-volatile solvent Vapour Pressure.

P_A = Solution Vapour Pressure.

x_B = Mole fraction of solute.

Vaht hoff factor $\Rightarrow \frac{P^{\circ}_A - P_A}{P^{\circ}_A} = x_B \cdot i$

$\Rightarrow$ Elevation in Boiling Point.

Addition of solute increases boiling point

$$\boxed{\Delta T_B = T_B - T_B^\circ}$$

| |
soln solvent

$\Delta T_B \propto \text{molality}$

$$\boxed{\Delta T_B = K_b \cdot m}$$

Vant hoff factor $\Rightarrow \boxed{\Delta T_B = K_b \cdot m \cdot i}$

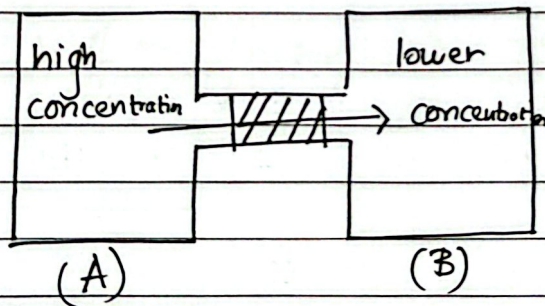
=> Depression in Freezing Point

$$\Delta T_f = T^{\circ}_f - T_f$$

$$\Delta T_f = k_f \cdot m$$

Vant hoff factor => $\Delta T_f = k_f \cdot m \cdot i$

=> Osmotic Pressure.



$$\pi = \frac{MRT}{0.0821}$$

molarity

in kelvin

$$\text{Molarity} = \frac{\text{moles of solute}}{\text{volm in litres}}$$

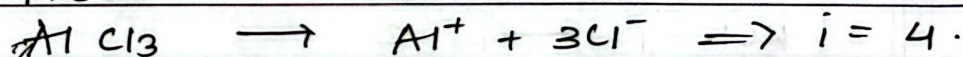
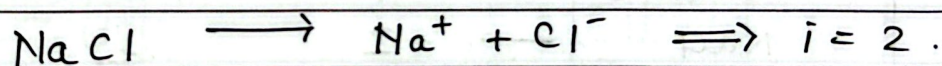
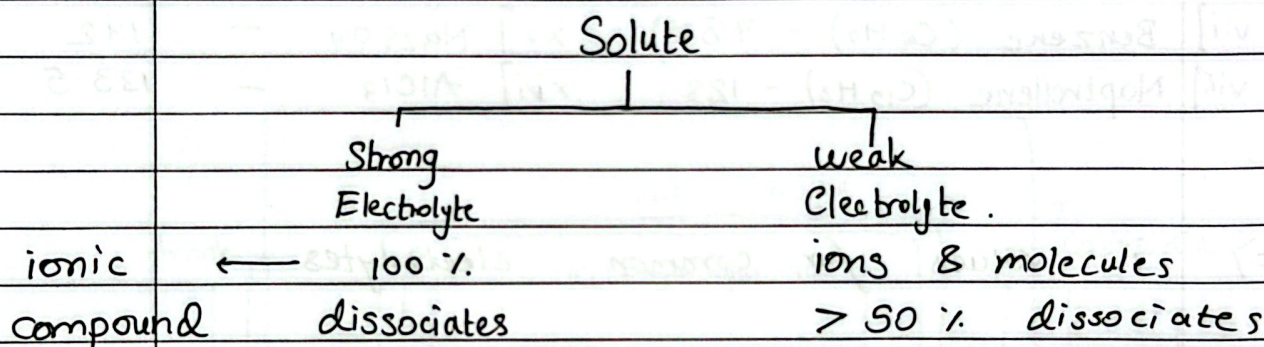
Vant hoff factor => $\pi = MRT \cdot i$

=> VANT HOFF FACTOR

When an electrolyte solution is taken it either

- Associates $\Rightarrow (i < 1)$
- Dissociates $\Rightarrow (i > 1)$

=> The amount by which it associates/dissociates is called the vant hoff factor (i)



Dissociation $\Rightarrow \quad \alpha = \frac{i-1}{n-1}$

Association $\Rightarrow \quad \alpha = \frac{i-1}{\frac{1}{n} - 1}$

=> $i = \frac{\text{Normal mol. mass}}{\text{calculated mol. mass}} \quad \text{or} \quad \frac{\text{observed colligative property}}{\text{normal colligative property}}$

or

$$i = \frac{\text{Total moles after asso/disso}}{\text{Total moles before asso/disso}}$$

IMPORTANT COMPOUNDS for Solutions Chapter.

=> Important compound masses.

i] Water (H_2O) — 18	ix] Camphor ($C_{10}H_{16}O$) — 152
ii] Glucose ($C_6H_{12}O_6$) — 180	x] Cyclohexane (C_6H_{12}) — 84
iii] Urea (CH_4N_2O) — 60	xi] NaCl ($NaCl$) — 58.5
iv] Sucrose ($C_{12}H_{22}O_{11}$) — 342	xii] KCl — 74.5
v] Acetic Acid (CH_3COOH) — 60	xiii] $CaCl_2$ — 111
vi] Ethanol (C_2H_5OH) — 46	xiv] $MgSO_4$ — 120
vii] Benzene (C_6H_6) — 78	xv] Na_2SO_4 — 142
viii] Napthalene ($C_{10}H_8$) — 128	xvi] $AlCl_3$ — 133.5

=> i values for common electrolytes

Glucose	—	1
NaCl	—	2
KCl	—	2
HCl	—	2
$MgSO_4$	—	2
$CaCl_2$	—	3
Na_2SO_4	—	3
K_2SO_4	—	3
$AlCl_3$	—	4